# Muhammad Kaif

AI Engineer • Machine Learning • Computer Vision • LLMs & AI Agents

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## PROFILE SUMMARY

AI Engineering undergraduate at FAST NUCES specialising in **machine learning, deep learning & computer vision, NLP & LLMs** and **knowledge representation & reasoning**. Builds models end to end, from raw data to deployment, with measurable results (**95.0% Recall@1** visual search, **$R^2$ 0.886** price prediction), and ships them as real products with LLM agents, FastAPI, full-stack apps and databases.

## EDUCATION

**National University of Computer and Emerging Sciences (FAST NUCES)** **Expected 2028**

*Bachelor of Science in Artificial Intelligence* *Islamabad, Pakistan*

Relevant coursework: Machine Learning, Knowledge Representation & Reasoning, Programming for AI, Data Structures, Databases, Operating Systems

## AI & TECHNICAL SKILLS

- **Machine Learning:** Supervised & unsupervised learning, feature engineering, evaluation, tuning, SHAP; Scikit-learn, XGBoost, Pandas
- **Deep Learning & Vision:** CNNs, transfer learning, object detection & tracking (YOLO11, ByteTrack), metric learning (triplet, ArcFace), image retrieval; OpenCV, MediaPipe, CLIP, FAISS
- **NLP & LLMs:** LLMs (Llama 3.3, Qwen), prompt engineering, RAG, speech-to-text (Whisper), vector databases (Pinecone)
- **Knowledge Representation & Reasoning:** Propositional & first-order logic, inference & resolution, knowledge graphs, A* search
- **AI Agents & Robotics:** LangChain, CrewAI, n8n, Vapi.ai voice agents, webhooks; ROS 2, Gazebo, ESP32, MQTT
- **Full Stack & Databases:** React, Next.js, Node.js, Express, FastAPI, GraphQL; PostgreSQL, MySQL, MongoDB, stored procedures
- **Languages & Tools:** Python, C++, JavaScript / TypeScript, SQL, Bash; Git, Linux, Docker, Jupyter, Postman

## AI & MACHINE LEARNING PROJECTS

**SoleMatch: Visual Sneaker Search Engine** *(ResNet50, Triplet Loss, CLIP, FAISS, FastAPI)* GitHub

- Fine-tuned ResNet50 with triplet loss over FAISS, lifting Recall@1 from 82.8% (zero-shot CLIP) to **95.0% on 645 held-out queries**.

**PitchVision AI: Football Match Analytics** *(YOLO11, ByteTrack, OpenCV, Homography, Streamlit)* GitHub

- Fine-tuned YOLO11 detectors (**ball found in 47% → 93% of frames**), ByteTrack, pitch homography, sprint & possession stats.

**House Price Prediction: End-to-End ML** *(Scikit-learn, XGBoost, SHAP, FastAPI, React)* GitHub

- Engineered ~93K listings, compared 3 models, tuned Random Forest to **$R^2$ 0.886**, explained it with SHAP and served it via FastAPI.

**CharaID: Open-Set Character Recognition** *(ArcFace, Metric Learning, FAISS)* GitHub

- Trained an ArcFace encoder (512-d embeddings) with a calibrated unknown threshold; new classes need 5 images, no retraining.

**Clara: AI Voice Receptionist** *(Vapi.ai, LLMs, n8n, Google Calendar)* GitHub

- Voice agent that answers clinic calls, checks live availability and books appointments through natural conversation.

**AI Automation Agents & Lead-Gen Pipelines** *(n8n, Groq Llama 3.3, Firecrawl, Tavily)* GitHub

- Autonomous LLM agents that scrape target sites, analyse business gaps and draft personalised outreach automatically.

**Synapse AI Wearable System** *(Whisper, Qwen, A* Search, ESP32)* GitHub

- Wearable sensor network combining Whisper speech-to-text, the Qwen LLM and A* pathfinding for context-aware AI assistance.

**Phantom-Key Touchless Interface** *(OpenCV, MediaPipe, Tkinter)* GitHub

- Real-time computer-vision controller mapping hand landmarks to mouse, keyboard, presentation and media commands.

## FULL STACK & SYSTEMS PROJECTS

- **RideFlow Ride-Sharing Platform**: Node.js, Express, MongoDB & SQL app with real-time matching, JWT auth, PKR wallet and SQL triggers. GitHub
- **Systems & 3D Web**: Movie Data Manager (C++ graph of 5,044 movies), Retail Pulse OS (POSIX multi-process pipeline), 3D Anime Chess (React Three Fiber). GitHub

## LEADERSHIP, ACTIVITIES & LANGUAGES

- **Team Lead, Civics Project** (FAST NUCES): organised the group, delegated tasks and presented the outcome; managed milestones on team AI projects.
- **Programming for AI Project**, FAST NUCES | **Member**, WOH Society | **Languages:** English (Fluent), Urdu (Native), German (Beginner)